

# **ECx00M&ECx00N&EC800K Series**

## **TCP/IP Application Note**

**LTE Standard Module Series**

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**Quectel Wireless Solutions Co., Ltd.**

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: [info@quectel.com](mailto:info@quectel.com)

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# About the Document

## Revision History

| Version | Date       | Author     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -       | 2020-07-24 | Larson LI  | Creation of the document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.0     | 2023-01-11 | Larson LI  | First official release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1.1     | 2023-01-13 | Larson LI  | Added the applicable module: EC800M-CN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.2     | 2023-10-30 | Fawei ZHOU | <ol style="list-style-type: none"> <li>Added the following applicable modules: EC800K series and EC600M-CN.</li> <li>Added AT+QICFG="wakeup/data" and URC +QIURC: "wakeup/data" (Chapter 2.3.1 &amp; Chapter 2.4.6).</li> <li>Added &lt;data_format&gt; and &lt;try_times&gt; in AT+QICFG="send/auto" (Chapter 2.3.1).</li> <li>Added &lt;data_format&gt; in AT+QICFG="recv/ignore" (Chapter 2.3.1).</li> <li>Added two notes of AT+QICFG (Chapter 2.3.1).</li> <li>Added an example of optional TCP configurations (Chapter 3.10).</li> </ol> |

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# 1 Introduction

Quectel LTE Standard ECx00M family, ECx00N family and EC800K series modules feature an embedded TCP/IP stack, which enables the host to access the Internet directly via AT commands. This significantly reduces the dependence on external PPP and TCP/IP protocol stacks and thus minimizes costs.

ECx00M family, ECx00N family and EC800K series modules provide socket services such as TCP client, UDP client, TCP server and UDP server.

## 1.1. Applicable Modules

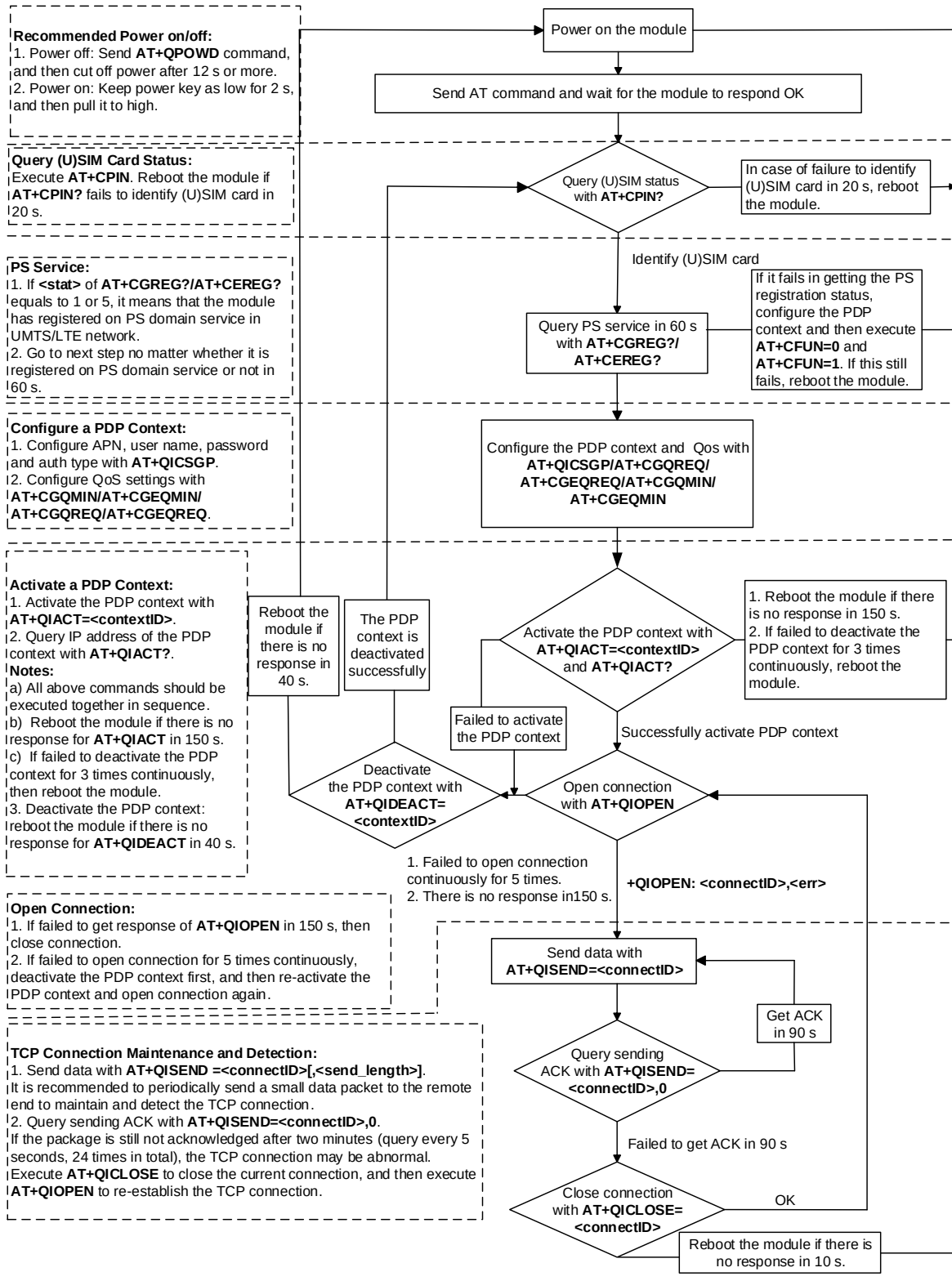
**Table 1: Applicable Modules**

| Module Family | Module        |
|---------------|---------------|
| ECx00M        | EC200M-CN     |
|               | EC600M-CN     |
|               | EC800M-CN     |
| ECx00N        | EC200N-CN     |
|               | EC800N-CN     |
| -             | EC800K Series |

## 1.2. The Process of Using TCP/IP AT Commands

Via TCP/IP AT commands, the host can configure a PDP context, activate/deactivate the PDP context, start/close a socket service and send/receive data via the socket service. The following figure illustrates how to use TCP/IP AT commands.





**Notes:**

- Please note that users need to wait for the final response (for example "OK", "CME ERROR", "CMS ERROR") of the last AT command you entered before you enter the next AT command. You can reboot the module if the module fails to get response in 60 s.
- Reboot the module if the module has not got response of **AT+QIACT** in 150 s or response of **AT+QICLOSE** in 10 s and in 40 s.
- It is NOT recommended to frequently reboot the module. When the module has been continuously rebooted for 3 times due to failed AT command execution, it can be rebooted immediately for the first time after that. If it still fails, reboot the module after 10 minutes for the second time, and reboot after 30 minutes for the third time, one hour for the fourth time, etc.

**Figure 1: Flow Chart of Using TCP/IP AT Commands**

### 1.3. Description of Data Access Modes

ECx00M family, ECx00N family and EC800K series modules support the following three kinds of data access modes:

- Buffer access mode
- Direct push mode
- Transparent transmission mode

When opening a socket service via **AT+QIOPEN**, the data access mode can be specified with **<access\_mode>**. After a socket service is opened, the access mode can be switched with **AT+QISWTMD**.

1. In buffer access mode, data can be sent via **AT+QISEND**. If the module has received the data from the Internet, it will buffer them and report the URC: **+QIURC: "recv",<connectID>**. The host can retrieve the buffered data with **AT+QIRD**.
2. In direct push mode, data can be sent via **AT+QISEND**. If the module has received the data from the Internet, it will output them to COM port directly through the URC:  
**+QIURC: "recv",<connectID>,<currentrecvlength><CR><LF><data>** or **+QIURC: "recv",<connectID>,<currentrecvlength>,<remoteIP>,<remote\_port><CR><LF><data>**.
3. In transparent transmission mode, the corresponding port (such as UART, USB modem port, etc.) is exclusively used for sending/receiving data directly to the Internet. The data received from COM port will be sent to the Internet directly, and the data received from Internet will be outputted via COM port directly.

#### ● Exit transparent transmission mode

To exit the transparent transmission mode either:

- 1) Execute **+++**. Follow the requirements below to prevent **+++** from being misinterpreted as data:
  - a) Do not input any character for at least 1 s before and after inputting **+++**.
  - b) Input only **+++** within 1 s, and wait until OK is returned. Once OK is returned, the module switches to buffer access mode.

OR

- 2) Change MAIN\_DTR from LOW to HIGH to make the module enter command mode. In this case, set **AT&D1** (see *document [1]*) before the module enters transparent transmission mode.

#### ● Return to transparent access mode

To return to transparent transmission mode either:

- 1) Execute **AT+QISWTMD**. Before execution specify the **<access\_mode>** as 2. Once transparent transmission mode is entered successfully, **CONNECT** is returned.

OR

- 2) Execute **ATO**. After a connection exits from transparent transmission mode, executing **ATO** switches the data access mode back to transparent transmission mode. Once transparent transmission mode is entered successfully, **CONNECT** is returned. If no connection has entered transparent transmission mode, **ATO** returns **NO CARRIER**.

**NOTE**

1. In buffer access mode, if the buffer is not empty, and the module receives data again, it does not report any new URC until all the received data have been retrieved with **AT+QIRD** from the buffer.
2. In transparent transmission mode, AT commands cannot be executed. If the socket connection is closed because of network error or other errors, the module reports **NO CARRIER** and exits the transparent transmission mode. In this case, execute **AT+QICLOSE** to close the socket service.

## 2 TCP/IP AT Commands

This chapter describes AT commands related to TCP/IP.

### 2.1. AT Command Introduction

#### 2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

#### 2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Type of AT Commands

| Command Type      | Syntax                                                          | Description                                                                                                          |
|-------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Test Command      | <b>AT+&lt;cmd&gt;=?</b>                                         | Test the existence of corresponding command and return information about the type, value, or range of its parameter. |
| Read Command      | <b>AT+&lt;cmd&gt;?</b>                                          | Check the current parameter value of a corresponding Write Command.                                                  |
| Write Command     | <b>AT+&lt;cmd&gt;=&lt;p1&gt;[,&lt;p2&gt;[,&lt;p3&gt;[...]]]</b> | Set user-definable parameter value.                                                                                  |
| Execution Command | <b>AT+&lt;cmd&gt;</b>                                           | Return a specific information parameter or perform a specific action.                                                |

## 2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you familiarize with AT commands and learn how to use them. The examples, however, should not be taken as Quectel's recommendation or suggestions about how you should design a program flow or what status you should set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there exists a correlation among these examples and that they should be executed in a given sequence.

## 2.3. AT Commands

### 2.3.1. AT+QICFG Configure Optional Parameters

This command configures optional parameters.

| AT+QICFG Configure Optional Parameters |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br>AT+QICFG=?             | Response<br>+QICFG: "transpktsize",(range of supported <transpktsize>s)<br>+QICFG: "transwaittm",(range of supported <transwaittm>s)<br>+QICFG: "dataformat",(list of supported <send_data_format>s),(list of supported <recv_data_format>s)<br>+QICFG: "viewmode",(list of supported <view_mode>s)<br>+QICFG: "passiveclosed",(list of supported <closed>s)<br>+QICFG: "udp/readmode",(list of supported <mode>s)<br>+QICFG: "udp/sendmode",(list of supported <mode>s)<br>+QICFG: "tcp/keepalive",(list of supported <enable>s),(range of supported <idle_time>s),(range of supported <interval_time>s),(range of supported <probe_cnt>s)<br>+QICFG: "recvind",(list of supported <show_length>s)<br>+QICFG: "tcp/retranscfg",(range of supported <retran_time>s),(range of supported <retran_time>s)<br>+QICFG: "tcp/accept",(list of supported <state>s)<br>+QICFG: "send/buffersize",(range of supported <buffer_size>s)<br>+QICFG: "send/auto",(range of supported <connectID>s),(range of supported <cycle_time>s),( <msg_auto>data sent),(range of supported <data_format>s),(range of supported <try_times>s)<br>+QICFG: "recv/ignore",(range of supported <connectID>s),(Filtered <msg_ignore> data),(range of supported <data_format>s)<br>+QICFG: "formatcfg",(range of supported <format>s)<br>+QICFG: "qisend/timeout",(range of supported <timeout>s) |

|                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                               | <p>+QICFG: "close/mode",(range of supported &lt;close_mode&gt;s)<br/> +QICFG: "sendinfo",(list of supported &lt;send_view_mode&gt;s)<br/> +QICFG: "TCP/SendMode",(list of supported &lt;send_mode&gt;s)<br/> +QICFG: "pdp/retry",(list of supported &lt;pdpmode&gt;s),(list of supported &lt;ratmode&gt;s),(range of supported &lt;counts&gt;s),(range of supported &lt;retry_time&gt;s)<br/> +QICFG: "wakeup/data",(range of supported &lt;connectID&gt;s),&lt;wakeup_data&gt;,(range of supported &lt;data_format&gt;s)</p> <p>OK</p> |
| <p>Write Command</p> <p>Query or set the packet size for transparent transmission mode</p> <p><b>AT+QICFG="transpktsize",&lt;transpktsize&gt;]</b></p>                                                        | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/> <b>+QICFG: "transpktsize",&lt;transpktsize&gt;</b></p> <p>OK</p> <p>If the optional parameter is specified, set the packet size for transparent transmission mode:<br/> OK</p> <p>If there is any error:<br/> <b>ERROR</b></p>                                                                                                                                                                                                                 |
| <p>Write Command</p> <p>Query or set the waiting time before automatically sending data in transparent transmission mode</p> <p><b>AT+QICFG="transwaittm",&lt;transwaittm&gt;]</b></p>                        | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/> <b>+QICFG: "transwaittm",&lt;transwaittm&gt;</b></p> <p>OK</p> <p>If the optional parameter is specified, set the waiting time before automatically sending data in transparent transmission mode:<br/> OK</p> <p>If there is any error:<br/> <b>ERROR</b></p>                                                                                                                                                                                 |
| <p>Write Command</p> <p>Query or set the format of the data to be sent or received (only for non-transparent mode)</p> <p><b>AT+QICFG="dataformat",&lt;send_data_format&gt;,&lt;recv_data_format&gt;]</b></p> | <p>Response</p> <p>If the optional parameters are omitted, query the current setting:<br/> <b>+QICFG: "dataformat",&lt;send_data_format&gt;,&lt;recv_data_format&gt;</b></p> <p>OK</p> <p>If the optional parameters are specified, set the format of the data to be sent or received:<br/> Ok</p> <p>If there is any error:</p>                                                                                                                                                                                                        |

|                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                   | <b>ERROR</b>                                                                                                                                                                                                                                                                                                                  |
| <p>Write Command</p> <p>Query or set the output format of received data (only for non-transparent mode)</p> <p><b>AT+QICFG="viewmode",&lt;view_mode&gt;</b></p>   | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "viewmode",&lt;view_mode&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the output format of the received data:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>              |
| <p>Write Command</p> <p>Enable or disable the passive close of TCP connection when the server is closed</p> <p><b>AT+QICFG="passiveclosed",&lt;closed&gt;</b></p> | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "passiveclosed",&lt;closed&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, enable or disable the passive close of TCP connection:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p> |
| <p>Write Command</p> <p>Query or set the mode when reading UDP data</p> <p><b>AT+QICFG="udp/readmode",&lt;mode&gt;</b></p>                                        | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "udp/readmode",&lt;mode&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the UDP data read mode:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>                               |
| <p>Write Command</p> <p>Query or set the mode when sending UDP data</p> <p><b>AT+QICFG="udp/sendmode",&lt;mode&gt;</b></p>                                        | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "udp/sendmode",&lt;mode&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the UDP data sending</p>                                                                                               |

|                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                       | <p>mode:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p>                                                                                                                                                                                                                                                                                                                    |
| <p>Write Command</p> <p>Query or set whether to send TCP keep-alive information.</p> <p><b>AT+QICFG="tcp/keepalive",&lt;enable&gt;,&lt;idle_time&gt;,&lt;interval_time&gt;,&lt;probe_cnt&gt;]</b></p> | <p>Response</p> <p>If the optional parameters are omitted, query the current configurations</p> <p><b>+QICFG: "tcp/keepalive",&lt;enable&gt;,&lt;idle_time&gt;,&lt;interval_time&gt;,&lt;probe_cnt&gt;]</b></p> <p><b>OK</b></p> <p>If the optional parameters are entered, set whether to send TCP keep-alive information:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p> |
| <p>Write Command</p> <p>Enable or disable data length presence in URC reported after data reception in buffer mode</p> <p><b>AT+QICFG="recvind",&lt;show_length&gt;]</b></p>                          | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:</p> <p><b>+QICFG: "recvind",&lt;show_length&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, enable or disable data length presence in URC reported:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p>                                                             |
| <p>Write Command</p> <p>Query or set the number of retransmissions and retransmission interval</p> <p><b>AT+QICFG="tcp/retranscfg",&lt;retran_times&gt;,&lt;retran_time&gt;]</b></p>                  | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:</p> <p><b>+QICFG: "tcp/retranscfg",&lt;retran_times&gt;,&lt;retran_time&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the number of retransmissions and retransmission interval:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p>                          |



|                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Write Command</p> <p>Query/enable or disable the automatic reception of the TCP connection from the client</p> <p><b>AT+QICFG="tcp/accept",&lt;state&gt;]</b></p>                                                | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "tcp/accept",&lt;state&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, enable or disable the automatic reception of the TCP connection from the client:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>                                             |
| <p>Write Command</p> <p>Query or set the maximum length of a single transmission data</p> <p><b>AT+QICFG="send/buffersize",&lt;buffer_size&gt;]</b></p>                                                             | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "send/buffersize",&lt;buffer_size&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the maximum length of a single transmission data:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>                                                             |
| <p>Write Command</p> <p>Query or set to send heartbeat packet data periodically</p> <p><b>AT+QICFG="send/auto",&lt;connectID&gt;,&lt;cycle_time&gt;,&lt;msg_auto&gt;,&lt;data_format&gt;,&lt;try_times&gt;]</b></p> | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "send/auto",&lt;connectID&gt;,&lt;cycle_time&gt;,&lt;msg_auto&gt;,&lt;data_format&gt;,&lt;try_times&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set to send heartbeat packet data periodically:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p> |
| <p>Write Command</p> <p>Query or set to filter the specified data</p> <p><b>AT+QICFG="recv/ignore",&lt;connectID&gt;,&lt;msg_ignore&gt;,&lt;data_format format&gt;]</b></p>                                         | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "recv/ignore",&lt;connectID&gt;,&lt;msg_ignore&gt;,&lt;data_format format format&gt;</b></p> <p><b>OK</b></p>                                                                                                                                                                                |

|                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                | <p>If the optional parameter is specified, set to filter the specified data:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>                                                                                                                                                                                |
| <p>Write Command</p> <p>Query or set the format of sending<br/><b>AT+QISEND</b> to return &gt;<br/><b>AT+QICFG="formatcfg",&lt;format&gt;</b><br/><b>&gt;]</b></p>             | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "formatcfg",&lt;format&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the output format of &gt;:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>                            |
| <p>Write Command</p> <p>Query or set the maximum response time for sending<br/><b>AT+QISEND</b><br/><b>AT+QICFG="qisend/timeout",&lt;timeout&gt;</b><br/><b>meout&gt;]</b></p> | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "qisend/timeout",&lt;timeout&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the timeout time after output &gt;:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p>             |
| <p>Write Command</p> <p>Query or set up asynchronous disconnection of TCP connection<br/><b>AT+QICFG="close/mode",&lt;close_mode&gt;</b><br/><b>e_mode&gt;]</b></p>            | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "close/mode",&lt;close_mode&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set up asynchronous disconnection of TCP connection:<br/><b>OK</b></p> <p>If there is any error:<br/><b>ERROR</b></p> |
| <p>Write Command</p> <p>Enable or disable displaying the return information of <b>AT+QISEND</b> or <b>AT+QISENDEX</b> in URC form</p>                                          | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:<br/><b>+QICFG: "sendinfo",&lt;send_view_mode&gt;</b></p>                                                                                                                                                                                 |

|                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>AT+QICFG="sendinfo"[,&lt;send_v<br/>ew_mode&gt;]</p>                                                                                                                                                 | <p><b>OK</b></p> <p>If the optional parameter is specified, enable or disable displaying the return information of <b>AT+QISEND</b> or <b>AT+QISENDEX</b> in URC form:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p>                                                                                                                                                                                                                |
| <p>Write Command</p> <p>Set the mode when sending <b>SEND</b></p> <p><b>OK</b></p> <p>AT+QICFG="TCP/SendMode"[,&lt;s<br/>end_mode&gt;]</p>                                                              | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:</p> <p><b>+QICFG: "TCP/SendMode",&lt;send_mode&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set the mode when sending <b>SEND</b> <b>OK</b>:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p>                                                                                                                           |
| <p>Write Command</p> <p>Configure retry times and time of PDP activation and deactivation</p> <p>AT+QICFG="pdp/retry",&lt;pdpmo<br/>de&gt;,&lt;ratmode&gt;[,&lt;counts&gt;,&lt;retry<br/>_time&gt;]</p> | <p>Response</p> <p>If the optional parameter is omitted, query the current setting:</p> <p><b>+QICFG: "pdp/retry",&lt;pdpmode&gt;,&lt;ratmode&gt;,&lt;counts&gt;,&lt;retry_time&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, configure retry times and time of PDP activation and deactivation:</p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p>                                                            |
| <p>Write Command</p> <p>Set wake-up packet data</p> <p>AT+QICFG="wakeup/data",&lt;con<br/>nectID&gt;[,&lt;wakeup_data&gt;[,&lt;data_<br/>format&gt;]]</p>                                               | <p>Response</p> <p>If the optional parameter is omitted, and <b>&lt;data_format&gt;</b> has not been configured, query the current setting:</p> <p><b>+QICFG: "wakeup/data",&lt;connectID&gt;,&lt;wakeup_data&gt;,0</b></p> <p><b>OK</b></p> <p>If the optional parameter is omitted, and <b>&lt;data_format&gt;</b> has been configured, query the current setting:</p> <p><b>+QICFG: "wakeup/data",&lt;connectID&gt;,&lt;wakeup_data&gt;,&lt;data_</b></p> |

|                       |                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <b>format&gt;</b><br><br><b>OK</b><br><br>If the optional parameter is specified, set wake-up packet data:<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b> |
| Maximum Response Time | 300 ms                                                                                                                                                                |
| Characteristic        | This command takes effect immediately.<br>The configurations are not saved.                                                                                           |

## Parameter

|                                 |                                                                                                                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;transpktsize&gt;</b>     | Integer type. In transparent transmission mode, the max length of the data packet to be sent. Range: 1–1460. Default value: 1024. Unit: byte.                                                                                                           |
| <b>&lt;transwaittm&gt;</b>      | Integer type. In transparent transmission mode, the waiting time before sending the data automatically, if the length of data to be sent is less than the specified value of <b>&lt;transpktsize&gt;</b> . Range: 0–20. Default value: 2. Unit: 100 ms. |
| <b>&lt;send_data_format&gt;</b> | Integer type. Format of the data to be sent. Suffix "0x" is not needed when the mode is set as Hex mode as the module will automatically form two bytes to one ASCII code.<br>0 Text mode<br>1 Hex mode                                                 |
| <b>&lt;recv_data_format&gt;</b> | Integer type. Format of the data to be received. Suffix "0x" is not needed when the mode is set as Hex mode as the module will automatically form two bytes to one ASCII code.<br>0 Text mode<br>1 Hex mode                                             |
| <b>&lt;view_mode&gt;</b>        | Integer type. Output format of received data.<br>0 data header\r\n\data.<br>1 data header,data.                                                                                                                                                         |
| <b>&lt;closed&gt;</b>           | Integer type. Enable or disable the passive close of TCP connection when the server is closed.<br>0 Disable<br>1 Enable                                                                                                                                 |
| <b>&lt;mode&gt;</b>             | Integer type. Mode when sending UDP data.<br>0 Disable block mode.<br>1 Enable stream mode.                                                                                                                                                             |
| <b>&lt;enable&gt;</b>           | Integer type. Enable or disable sending of TCP keep-alive information.<br>0 Disable                                                                                                                                                                     |

|                  |                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------|
|                  | 1    Enable                                                                                                              |
| <idle_time>      | Integer type. Indicates the cycle time of triggered keepalive. Range: 1–1800. Unit: s.                                   |
| <interval_time>  | Integer type. Indicates the interval of sending a packet in cycle time. Range: 25–100. Unit: s.                          |
| <probe_cnt>      | Integer type. Packet transmission count in cycle time. Range: 3–10.                                                      |
| <show_length>    | Integer type. Enable or disable data length presence in URC reported after data reception in TCP/IP buffer mode.         |
|                  | 0    Disable                                                                                                             |
|                  | 1    Enable                                                                                                              |
| <retran_times>   | Integer type. Number of reconnections within cycle time. Range: 3–12.                                                    |
| <retran_time>    | Integer type. Reconnect interval within cycle time. Range: 5–1000. Unit: ms.                                             |
| <state>          | Integer type.                                                                                                            |
|                  | 0    Disable auto accepting incoming TCP connection from the client.                                                     |
|                  | 1    Enable auto accepting incoming TCP connection from the client.                                                      |
| <buffer_size>    | Integer type. Maximum number of bytes sent at one time. Range: 1460–10240.                                               |
| <connectID>      | Integer type. Socket service index. Range: 0–11.                                                                         |
| <cycle_time>     | Integer type. Set the interval for sending heartbeat packets. Range: 20–86400. Unit: s.                                  |
| <msg_auto>       | String type. Send the content of the heartbeat packet.                                                                   |
| <msg_ignore>     | String type. Filter data content.                                                                                        |
| <format>         | Integer type. Control > output format.                                                                                   |
|                  | 0    0D0A>                                                                                                               |
|                  | 1    0D0A>0D0A                                                                                                           |
| <timeout>        | Integer type. Send data timeout time. Range: 0–120. Unit: s.                                                             |
| <close_mode>     | Integer type. Enable or disable asynchronous disconnection of TCP connection.                                            |
|                  | 0    Disable                                                                                                             |
|                  | 1    Enable                                                                                                              |
| <send_view_mode> | Integer type. Enable or disable displaying the return information of <b>AT+QISEND</b> or <b>AT+QISENDEX</b> in URC form. |
|                  | 0    Disable                                                                                                             |
|                  | 1    Enable                                                                                                              |
| <send_mode>      | Integer type. The return mode of <b>SEND OK</b> .                                                                        |
|                  | 0    Return <b>SEND OK</b> immediately after sending data                                                                |
|                  | 1    Return <b>SEND OK</b> after receiving ACK from server                                                               |
|                  | 2    Return <connectID>, <b>SEND OK</b> after receiving ACK from server                                                  |
|                  | 3    Return <b>+QIURC: SEND OK</b> after receiving ACK from server                                                       |
|                  | 4    Return <b>+QIURC: &lt;connectID&gt;, SEND OK</b> after receiving ACK from server                                    |
| <pdpmode>        | Integer type. Set the PDP activation or deactivation mode.                                                               |
|                  | 0    Deactivation mode                                                                                                   |
|                  | 1    Activation mode                                                                                                     |

|                            |                                                                                                                                                                        |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;ratmode&gt;</b>     | Integer type. Set the network type.<br>0 4G mode<br>1 2G mode                                                                                                          |
| <b>&lt;counts&gt;</b>      | Integer type. Number of retry times. Range: 2–4. Default value: 4. The actual number of activation or deactivation is equal to the value of <b>&lt;counts&gt;</b> + 1. |
| <b>&lt;retry_time&gt;</b>  | Integer type. The maximum time for a single activation and deactivation.<br>Unit: s.                                                                                   |
| <b>&lt;wakeup_data&gt;</b> | String type. Wake-up packet data.                                                                                                                                      |
| <b>&lt;data_format&gt;</b> | Integer type. The type of the input packet data.<br>0 Hexadecimal ASCII code (automatically converted to hexadecimal string)<br>1 Hexadecimal string<br>2 String       |
| <b>&lt;try_times&gt;</b>   | Integer type. Retry times. Range: 0–10.                                                                                                                                |

#### NOTE

1. If **<data\_format>**=2, that is, the input data is a string type, the string data cannot contain special characters (such as ASCII special control characters and escape characters), otherwise **<data\_format>** should be configured to 0 or 1.
2. After the specified data is configured with **AT+QICFG="wakeup/data"**, all other data sent by the server will be discarded, and the MCU will be notified to wake up only when the packet is awakened or an abnormal event occurs in the network.

### 2.3.2. AT+QICSGP Configure Parameters of a TCP/IP Context

This command configures the **<APN>**, **<username>**, **<password>** and other parameters of a TCP/IP context. The QoS settings can be configured with **AT+CGQMIN**, **AT+CGEQMIN**, **AT+CGQREQ** and **AT+CGEQREQ**. For more information about the AT commands, see *document [1]*.

#### AT+QICSGP Configure Parameters of a TCP/IP Context

|                                                                                               |                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QICSGP=?</b>                                                            | Response<br><b>+QICSGP:</b> (range of supported <b>&lt;contextID&gt;</b> s),(range of supported <b>&lt;context_type&gt;</b> s), <b>&lt;APN&gt;</b> , <b>&lt;username&gt;</b> , <b>&lt;password&gt;</b> ,(range of supported <b>&lt;authentication&gt;</b> s),(list of supported <b>&lt;CDMA_pwd&gt;</b> s)<br><br>OK |
| Write Command<br>Query a specific context configuration<br><b>AT+QICSGP=&lt;contextID&gt;</b> | Response<br><b>+QICSGP:</b> <b>&lt;context_type&gt;</b> , <b>&lt;APN&gt;</b> , <b>&lt;username&gt;</b> , <b>&lt;password&gt;</b> , <b>&lt;authentication&gt;</b><br><br>OK                                                                                                                                           |

|                                                                                                                                                                                                 |                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Write Command<br>Configure the context<br><b>AT+QICSGP=&lt;contextID&gt;[,&lt;context_type&gt;,&lt;APN&gt;[,&lt;username&gt;,&lt;password&gt;][,&lt;authentication&gt;[,&lt;CDMA_pwd&gt;]]]</b> | Response<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b>         |
| Maximum Response Time                                                                                                                                                                           | /                                                                           |
| Characteristic                                                                                                                                                                                  | This command takes effect immediately.<br>The configurations are not saved. |

## Parameter

|                               |                                                                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;contextID&gt;</b>      | Integer type. Context ID. Range: 1–15.                                                                                                                           |
| <b>&lt;context_type&gt;</b>   | Integer type. Protocol type.<br><u>1</u> IPv4<br>2    IPv6<br>3    IPv4v6                                                                                        |
| <b>&lt;APN&gt;</b>            | String type. Access point name.                                                                                                                                  |
| <b>&lt;username&gt;</b>       | String type. Username. Maximum length: 127 bytes.                                                                                                                |
| <b>&lt;password&gt;</b>       | String type. Password. Maximum length: 127 bytes.                                                                                                                |
| <b>&lt;authentication&gt;</b> | Integer type. Authentication methods.<br><u>0</u> None<br>1    PAP<br>2    CHAP<br>3    PAP or CHAP                                                              |
| <b>&lt;CDMA_pwd&gt;</b>       | Integer type. To configure whether to enable saving of <b>&lt;username&gt;</b> and <b>&lt;password&gt;</b> over CDMA network.<br><u>0</u> Disable<br>1    Enable |

## Example

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| <b>AT+QICSGP=1</b>                      | //Query the configuration of context 1.            |
| <b>+QICSGP: 1,"", "", "", 0</b>         |                                                    |
| <b>OK</b>                               |                                                    |
| <b>AT+QICSGP=1,1,"UNINET","", "", 1</b> | //Configure context 1. China Unicom APN: "UNINET". |
| <b>OK</b>                               |                                                    |

### 2.3.3. AT+QIACT Activate a PDP Context

Before activating a PDP context with **AT+QIACT**, the context should be configured with **AT+QICSGP**. After activation, the IP address can be queried with **AT+QIACT?**.

Although the range of **<contextID>** is 1–15, the module supports maximum three PDP contexts activated simultaneously. Depending on the network, it may take max. 150 s to return **OK** or **ERROR** after executing **AT+QIACT**. Other AT commands can be executed only after the response is returned.

#### AT+QIACT Activate a PDP Context

|                                                                                        |                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QIACT=?</b>                                                      | Response<br><b>+QIACT:</b> (range of supported <b>&lt;contextID&gt;</b> s)<br><br><b>OK</b>                                                                                                                                                                                                             |
| Read command<br><b>AT+QIACT?</b>                                                       | Response<br>Return the list of the currently activated contexts and their IP addresses:<br><b>+QIACT: 1,&lt;context_state&gt;,&lt;context_type&gt;[,&lt;IP_addresses&gt;]</b><br><b>[.....</b><br><b>+QIACT: 15,&lt;context_state&gt;,&lt;context_type&gt;[,&lt;IP_addresses&gt;]]</b><br><br><b>OK</b> |
| Write Command<br>Activate a specified PDP context<br><b>AT+QIACT=&lt;contextID&gt;</b> | Response<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b>                                                                                                                                                                                                                                     |
| Maximum Response Time                                                                  | 150 s, determined by the network.                                                                                                                                                                                                                                                                       |
| Characteristic                                                                         | /                                                                                                                                                                                                                                                                                                       |

#### Parameter

|                              |                                                              |
|------------------------------|--------------------------------------------------------------|
| <b>&lt;contextID&gt;</b>     | Integer type. Context ID. Range: 1–15.                       |
| <b>&lt;context_state&gt;</b> | Integer type. Context state.<br>0 Deactivated<br>1 Activated |
| <b>&lt;context_type&gt;</b>  | Integer type. Protocol type.<br>1 IPv4<br>2 IPv6             |



|              |                                                               |        |
|--------------|---------------------------------------------------------------|--------|
|              | 3                                                             | IPv4v6 |
| <IP_address> | String type. Local IP address after the context is activated. |        |

### 2.3.4. AT+QIDEACT Deactivate a PDP Context

This command deactivates a specific context and closes all TCP/IP connections set up in this context. Depending on the network, it may take max. 40 s to return **OK** or **ERROR** after executing **AT+QIDEACT**. Other AT commands can be executed only after the response is returned.

| AT+QIDEACT Deactivate a PDP Context                  |                                                                                 |
|------------------------------------------------------|---------------------------------------------------------------------------------|
| Test Command<br><b>AT+QIDEACT=?</b>                  | Response<br><b>+QIDEACT:</b> (range of supported <contextID>s)<br><br><b>OK</b> |
| Write Command<br><b>AT+QIDEACT=&lt;contextID&gt;</b> | Response<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b>             |
| Maximum Response Time                                | 40 s, determined by the network.                                                |
| Characteristic                                       | /                                                                               |

#### Parameter

|             |                                        |
|-------------|----------------------------------------|
| <contextID> | Integer type. Context ID. Range: 1–15. |
|-------------|----------------------------------------|

### 2.3.5. AT+QIOPEN Open a Socket Service

The command opens a socket service. The service type can be specified by <service\_type>. The data access mode (buffer access mode, direct push mode and transparent transmission mode) can be specified by <access\_mode>. The URC **+QIOPEN** indicates if the socket service has been opened successfully.

1. If <service\_type> is "TCP LISTENER", the module works as TCP server. After accepting a new TCP connection, the module automatically specifies a <connectID> and reports the URC **+QIURC: "incoming",<connectID>,<serverID>,<remoteIP>,<remote\_port>**. The range of <connectID> is 0–11. The type of this new incoming connection is "TCP INCOMING" and the <access\_mode> of "TCP INCOMING" is the same as that of "TCP LISTENER".
2. If <service\_type> is "UDP SERVICE", UDP data can be sent to or received from the remote IP via <local\_port>.

- Send data: execute **AT+QISEND=<connectID>,<send\_length>,<remoteIP>,<remote\_port>**.
- Receive data in direct push mode: the module reports the URC **+QIURC: "recv",<connectID>,<currentrecvlength>,<remoteIP>,<remote\_port><CR><LF><data>**.
- Receive data in buffer access mode: the module reports the URC **+QIURC: "recv",<connectID>**, and then the received data can be retrieved with **AT+QIRD=<connectID>**.

3. It is suggested to wait for 150 s for **+QIOPEN: <connectID>,<err>** to be outputted after executing the Write Command. If the response cannot be received in 150 s, use **AT+QICLOSE** to close the socket.

## AT+QIOPEN Open a Socket Service

|                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QIOPEN=?</b>                                                                                                                                                                 | Response<br><b>+QIOPEN:</b> (range of supported <contextID>s),(range of supported <connectID>s),"TCP/UDP/TCP LISTENER/UDP SERVICE", "<IP_address>/<domain_name>", (range of supported <remote_port>s),(range of supported <local_port>s),(range of supported <access_mode>s)<br><br><b>OK</b>                                                                                                                                                                                                                        |
| Write Command<br><b>AT+QIOPEN=&lt;contextID&gt;,&lt;connectID&gt;,&lt;service_type&gt;,"&lt;IP_address&gt;/&lt;domain_name&gt;",&lt;remote_port&gt;[,&lt;local_port&gt;[,&lt;access_mode&gt;]]</b> | Response<br>If the service is in transparent transmission mode (<access_mode>=2) and the service is opened successfully:<br><b>CONNECT</b><br><br>If there is any error:<br><b>ERROR</b><br>Error description can be got via <b>AT+QIGETERROR</b> .<br><br>If the service is in buffer access mode (<access_mode>=0) or direct push mode (<access_mode>=1):<br><b>OK</b><br><br><b>+QIOPEN: &lt;connectID&gt;,&lt;err&gt;</b><br><err> is 0 when the service is opened successfully. In other cases, <err> is not 0. |
| Maximum Response Time                                                                                                                                                                              | 150 s, determined by the network.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Characteristic                                                                                                                                                                                     | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Parameter

|                |                                                  |
|----------------|--------------------------------------------------|
| <contextID>    | Integer type. Context ID. Range: 1–15.           |
| <connectID>    | Integer type. Socket service index. Range: 0–11. |
| <service_type> | String type. Socket service type.                |

|               |                                                                                                                                                                                                                                                                                                                                                                                  |                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|               | "TCP"                                                                                                                                                                                                                                                                                                                                                                            | Start a TCP connection as a client                       |
|               | "UDP"                                                                                                                                                                                                                                                                                                                                                                            | Start a UDP connection as a client                       |
|               | "TCP LISTENER"                                                                                                                                                                                                                                                                                                                                                                   | Start a TCP server to listen to incoming TCP connections |
|               | "UDP SERVICE"                                                                                                                                                                                                                                                                                                                                                                    | Start a UDP service                                      |
| <IP_address>  | String type.<br>If <service_type> is "TCP" or "UDP", it indicates the IP address of remote server, such as 220.180.239.212.<br>If <service_type> is "TCP LISTENER" or "UDP SERVICE", enter 127.0.0.1.                                                                                                                                                                            |                                                          |
| <domain_name> | String type. Domain name address of the remote server.                                                                                                                                                                                                                                                                                                                           |                                                          |
| <remote_port> | Integer type. Port number of the remote server. Range: 0–65535.<br>If <service_type> is "TCP" or "UDP", this parameter must be specified.<br>Range: 1–65535.<br>If <service_type> is "TCP LISTENER" or "UDP SERVICE", specify this parameter as 0.                                                                                                                               |                                                          |
| <local_port>  | Integer type. Local port number. Range: 0–65535.<br>If <service_type> is "TCP LISTENER" or "UDP SERVICE", this parameter must be specified. Range: 1–65535.<br>If <service_type> is "TCP" or "UDP", this parameter can be omitted and the default value is 0. The local port is assigned automatically if <local_port> is 0. Otherwise, the local port is assigned as specified. |                                                          |
| <access_mode> | Integer type. Data access mode of the socket service.<br>0 Buffer access mode<br>1 Direct push mode<br>2 Transparent transmission mode                                                                                                                                                                                                                                           |                                                          |
| <err>         | Integer type. Error codes of the operation. See <b>Chapter 4</b> for more information.                                                                                                                                                                                                                                                                                           |                                                          |

### 2.3.6. AT+QICLOSE Close a Socket Service

This command closes a specified socket service. Depending on the network, it will take max. 10 s (default value, can be modified by <timeout>) to return **OK** or **ERROR** after executing **AT+QICLOSE**. Other AT commands can be executed only after the response is returned.

| AT+QICLOSE Close a Socket Service                                      |                                                                                                                                  |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QICLOSE=?</b>                                    | Response<br><b>+QICLOSE:</b> (range of supported <connectID>s),(range of supported <timeout>s)<br><br><b>OK</b>                  |
| Write Command<br><b>AT+QICLOSE=&lt;connectID&gt;[,&lt;timeout&gt;]</b> | Response<br>If the socket service is closed successfully:<br><b>OK</b><br><br>If the command failed to close the socket service: |

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
|                       | <b>ERROR</b>                                                           |
| Maximum Response Time | 10 s by default, determined by the setting of <b>&lt;timeout&gt;</b> . |
| Characteristic        | /                                                                      |

## Parameter

|                          |                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;connectID&gt;</b> | Integer type. Socket service index. Range: 0–11.                                                                                                                                                                                                 |
| <b>&lt;timeout&gt;</b>   | Integer type. Timeout value for the response to be outputted. If the <b>FIN ACK</b> of the other peer is not received within <b>&lt;timeout&gt;</b> , the module will be forced to close the socket. Range: 0–65535. Default value: 10. Unit: s. |

### 2.3.7. AT+QISTATE Query Socket Service Status

The command queries the socket service status. If the **<query\_type>** is 0, it returns the statuses of all existing socket services in the specified context. If the **<query\_type>** is 1, it will return the status of a specified socket service.

| <b>AT+QISTATE Query Socket Service Status</b>                                                                                                                    |                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QISTATE=?</b>                                                                                                                              | Response<br><b>OK</b>                                                                                                                                                                                                                                                                                                        |
| Read/Execution Command<br><b>AT+QISTATE?</b> or <b>AT+QISTATE</b>                                                                                                | Response<br>Return the status of all existing connections:<br><b>+QISTATE: &lt;connectID&gt;,&lt;service_type&gt;,&lt;IP_address&gt;,&lt;remote_port&gt;,&lt;local_port&gt;,&lt;socket_state&gt;,&lt;contextID&gt;,&lt;serverID&gt;,&lt;access_mode&gt;,&lt;AT_port&gt;</b><br>[...]<br><br><b>OK</b>                        |
| Write Command<br>If <b>&lt;query_type&gt;</b> is 0, query the connection status of a specified context<br><b>AT+QISTATE=&lt;query_type&gt;,&lt;contextID&gt;</b> | Response<br>Return the status of all existing connections in a specified context:<br><b>+QISTATE: &lt;connectID&gt;,&lt;service_type&gt;,&lt;IP_address&gt;,&lt;remote_port&gt;,&lt;local_port&gt;,&lt;socket_state&gt;,&lt;contextID&gt;,&lt;serverID&gt;,&lt;access_mode&gt;,&lt;AT_port&gt;</b><br>[...]<br><br><b>OK</b> |
| Write Command<br>If <b>&lt;query_type&gt;</b> is 1, query the connection status of a specified socket                                                            | Response<br><b>+QISTATE: &lt;connectID&gt;,&lt;service_type&gt;,&lt;IP_address&gt;,&lt;remote_port&gt;,&lt;local_port&gt;,&lt;socket_state&gt;,&lt;contextID&gt;,&lt;</b>                                                                                                                                                    |

|                                                                |                                                         |
|----------------------------------------------------------------|---------------------------------------------------------|
| service <b>AT+QISTATE=&lt;query_type&gt;,&lt;connectID&gt;</b> | <b>serverID&gt;,&lt;access_mode&gt;,&lt;AT_port&gt;</b> |
|                                                                | <b>OK</b>                                               |
| Maximum Response Time                                          | 300 ms                                                  |
| Characteristic                                                 | /                                                       |

## Parameter

|                             |                                                                                                                                                                                                                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;query_type&gt;</b>   | Integer type. Query type.<br>0 Query connection status of all socket services in a specified context<br>1 Query connection status of a specified socket service                                                                                                                                                |
| <b>&lt;contextID&gt;</b>    | Integer type. Context ID. Range: 1–15.                                                                                                                                                                                                                                                                         |
| <b>&lt;connectID&gt;</b>    | Integer type. Socket service index. Range: 0–11.                                                                                                                                                                                                                                                               |
| <b>&lt;service_type&gt;</b> | String type. Socket service type.<br>"TCP" Start a TCP connection as a client<br>"UDP" Start a UDP connection as a client<br>"TCP LISTENER" Start a TCP server to listen to incoming TCP connections<br>"TCP INCOMING" Start a TCP connection accepted by a TCP server<br>"UDP SERVICE" Start a UDP service    |
| <b>&lt;IP_address&gt;</b>   | String type. IP address.<br>If <b>&lt;service_type&gt;</b> ="TCP" or "UDP", it is the IP address of remote server.<br>If <b>&lt;service_type&gt;</b> ="TCP LISTENER" or "UDP SERVICE", it is the local IP address.<br>If <b>&lt;service_type&gt;</b> ="TCP INCOMING", it is the IP address of a remote client. |
| <b>&lt;remote_port&gt;</b>  | Integer type. Remote port number.<br>If <b>&lt;service_type&gt;</b> ="TCP" or "UDP", it is the port of remote server.<br>If <b>&lt;service_type&gt;</b> ="TCP LISTENER" or "UDP SERVICE", the port is invalid.<br>If <b>&lt;service_type&gt;</b> ="TCP INCOMING", it is the port of a remote client.           |
| <b>&lt;local_port&gt;</b>   | Integer type. Local port number.<br>If <b>&lt;local_port&gt;</b> =0, then the local port is assigned automatically.                                                                                                                                                                                            |
| <b>&lt;socket_state&gt;</b> | Integer type. Socket service status.<br>0 "Initial": connection has not been established<br>1 "Opening": client is connecting or server is trying to listen<br>2 "Connected": client/incoming connection has been established<br>3 "Listening": server is listening<br>4 "Closing": connection is closing      |
| <b>&lt;serverID&gt;</b>     | Integer type. Only valid when <b>&lt;service_type&gt;</b> is "TCP INCOMING". <b>&lt;serverID&gt;</b> represents the ID of the server that accepts the incoming TCP connection, and the value is the same as <b>&lt;connectID&gt;</b> of the server's "TCP LISTENER".                                           |
| <b>&lt;access_mode&gt;</b>  | Integer type. Data access mode.<br>0 Buffer access mode<br>1 Direct push mode<br>2 Transparent transmission mode                                                                                                                                                                                               |

|            |                                          |  |
|------------|------------------------------------------|--|
| <AT_port>  | String type. COM port of socket service. |  |
| "usbmodem" | USB modem port                           |  |
| "usbat"    | USB AT port                              |  |
| "uart1"    | UART port1                               |  |
| "cmux1"    | MUX port 1                               |  |
| "cmux2"    | MUX port 2                               |  |
| "cmux3"    | MUX port 3                               |  |
| "cmux4"    | MUX port 4                               |  |

### 2.3.8. AT+QISEND Send Data

In buffer access mode (<access\_mode>=0) or direct push mode (<access\_mode>=1), the data can be sent with **AT+QISEND**. If the data have been sent to the module successfully, **SEND OK** is returned, otherwise **SEND FAIL** or **ERROR** is returned.

- **SEND FAIL** indicates the sending buffer is full. In this case, resending of the data can be tried.
- **ERROR** indicates an error in the sending data process. In this case, wait for some time before resending the data. Maximum length of data to be sent: 1460 bytes.
- **SEND OK** means that the data have been sent to the peer, but it does not mean they have reached the server successfully. You can query whether the data have reached the server with **AT+QISEND=<connectID>,0**.

| AT+QISEND Send Data                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QISEND=?</b>                                                                                                     | Response<br><b>+QISEND:</b> (range of supported <connectID>s),(range of supported <send_length>s)<br><br><b>OK</b>                                                                                                                                                                                                                                                                                                                                       |
| Write Command<br>Send variable-length data when <service_type> is "TCP", "UDP" or "TCP INCOMING"<br><b>AT+QISEND=&lt;connectID&gt;</b> | Response<br>><br>After the response >, input the data to be sent. Tap <b>Ctrl + Z</b> to send data, and tap <b>Esc</b> to cancel the operation.<br><br>(1) If <send_view_mode>=0:<br>If the connection has been established and the data is sent successfully:<br><b>SEND OK</b><br><br>If the connection has been established but the sending buffer is full:<br><b>SEND FAIL</b><br><br>If the connection has not been established, abnormally closed, |

|                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                     | <p>or any parameter is incorrect:<br/><b>ERROR</b></p> <p>(2) If <b>&lt;send_view_mode&gt;=1</b>:<br/>If the connection has been established and the data is sent successfully:<br/><b>+QISEND: &lt;connectID&gt;,&lt;status&gt;,&lt;Freesize&gt;</b></p> <p><b>OK</b></p> <p>If the connection has not been established, abnormally closed, or any parameter is incorrect:<br/><b>ERROR</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>Write Command</p> <p>Send fixed-length data when <b>&lt;service_type&gt;</b> is "TCP", "UDP" or "TCP INCOMING"</p> <p><b>AT+QISEND=&lt;connectID&gt;,&lt;send_length&gt;</b></p> | <p>Response</p> <p><b>&gt;</b></p> <p>After the response <b>&gt;</b>, input the data until the data length equals to <b>&lt;send_length&gt;</b>.</p> <p>(1) If <b>&lt;send_view_mode&gt;=0</b>:<br/>If the connection has been established and the data is sent successfully:<br/><b>SEND OK</b></p> <p>If the connection has been established but the sending buffer is full:<br/><b>SEND FAIL</b></p> <p>If the connection has not been established, abnormally closed, or any parameter is incorrect:<br/><b>ERROR</b></p> <p>(2) If <b>&lt;send_view_mode&gt;=1</b>:<br/>If the connection has been established:<br/><b>+QISEND: &lt;connectID&gt;,&lt;status&gt;,&lt;Freesize&gt;</b></p> <p><b>OK</b></p> <p>If the connection has not been established, abnormally closed, or any parameter is incorrect:<br/><b>ERROR</b></p> |
| <p>Write Command</p> <p>If <b>&lt;service_type&gt;</b> is "UDP SERVICE"</p> <p><b>AT+QISEND=&lt;connectID&gt;,&lt;send_length&gt;</b></p>                                           | <p>Response</p> <p>Send fixed length data to a specified remote IP address and remote port. The <b>&lt;service_type&gt;</b> must be "UDP SERVICE".</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gth>,<remoteIP>,<remote_port>                                                                                              | <p>&gt;</p> <p>After the response &gt;, input the data until the data length equals &lt;send_length&gt;</p> <p>If the connection is established and the data are sent successfully:</p> <p><b>SEND OK</b></p> <p>If the connection is established but the sending buffer is full:</p> <p><b>SEND FAIL</b></p> <p>If the connection is not established, abnormally closed, or any parameter is incorrect:</p> <p><b>ERROR</b></p> |
| <p>Write Command</p> <p>When &lt;send_length&gt; is 0, query the sent data</p> <p><b>AT+QISEND=&lt;connectID&gt;,0</b></p> | <p>Response</p> <p>If the specified connection exists:</p> <p><b>+QISEND: &lt;total_send_length&gt;,&lt;ackedbytes&gt;,&lt;unacked bytes&gt;</b></p> <p><b>OK</b></p> <p>If the specified &lt;connectID&gt; does not exist, or there is any other error:</p> <p><b>ERROR</b></p>                                                                                                                                                 |
| Maximum Response Time                                                                                                      | /                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Characteristic                                                                                                             | /                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Parameter

|                  |                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------|
| <connectID>      | Integer type. Socket service index. Range: 0–11.                                                        |
| <send_view_mode> | Integer type. Enable or disable display <b>AT+QISEND</b> command execution information in URC form.     |
|                  | <p>0 Disable</p> <p>1 Enable</p>                                                                        |
| <status>         | Integer type.                                                                                           |
|                  | <p>0 Send data to the socket buffer successfully</p> <p>1 The receiving buffer is full, send failed</p> |
| <Freesize>       | Integer type. Free space in the current buffer. Range: 0–10240. Unit: byte.                             |
| <send_length>    | Integer type. Length of data to be sent. Range: 0–1460. Unit: byte.                                     |
| <remoteIP>       | String type. Remote IP address (must be dot format). Valid only when <service_type> is "UDP SERVICE".   |
| <remote_port>    | Integer type. Remote port. Valid only when <service_type> is "UDP SEVERVICE".                           |



|                                  |                                                                |
|----------------------------------|----------------------------------------------------------------|
| <b>&lt;total_send_length&gt;</b> | Integer type. Total length of sent data. Unit: byte.           |
| <b>&lt;ackedbytes&gt;</b>        | Integer type. Total length of acknowledged data. Unit: byte.   |
| <b>&lt;unackedbytes&gt;</b>      | Integer type. Total length of unacknowledged data. Unit: byte. |

### 2.3.9. AT+QIRD Read the Received TCP/IP Data

In buffer access mode, after receiving data, the module will buffer it and report **+QIURC:** "recv",<connectID>, and then the data can be read by **AT+QIRD**.

Please note that if the buffer is not empty, and the module receives data again, it will not report a new URC until all the received data has been read via **AT+QIRD** from buffer.

| AT+QIRD Read the Received TCP/IP Data                                                                                          |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QIRD=?</b>                                                                                               | Response<br><b>+QIRD:</b> (range of supported <connectID>s),(range of supported <read_length>s)<br><br><b>OK</b>                                                                                                                                                                                |
| Write Command<br>When <service_type> is "TCP", "UDP", "TCP INCOMING"<br><b>AT+QIRD=&lt;connectID&gt;[,&lt;read_length&gt;]</b> | Response<br>If the specified connection has received the data, response:<br><b>+QIRD: &lt;read_actual_length&gt;&lt;CR&gt;&lt;LF&gt;&lt;data&gt;</b><br><br><b>OK</b><br><br>If there are no data:<br><b>+QIRD: 0</b><br><br><b>OK</b><br><br>If the connection does not exist:<br><b>ERROR</b> |
| Write Command<br>When <service_type> is "UDP SERVICE"<br><b>AT+QIRD=&lt;connectID&gt;</b>                                      | Response<br>If data exist:<br><b>+QIRD: &lt;read_actual_length&gt;,&lt;remoteIP&gt;,&lt;remote_port&gt;&lt;CR&gt;&lt;LF&gt;&lt;data&gt;</b><br><br><b>OK</b><br><br>If there are no data:<br><b>+QIRD: 0</b><br><br><b>OK</b>                                                                   |

|                                                                                                                                             |                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                             | <p>If the connection does not exist:</p> <p><b>ERROR</b></p>                                                                                                                                                                   |
| <p>Write Command</p> <p>When <b>&lt;read_length&gt;</b> is 0, query the retrieved data length</p> <p><b>AT+QIRD=&lt;connectID&gt;,0</b></p> | <p>Response</p> <p>If the specified connection exists:</p> <p><b>+QIRD: &lt;total_receive_length&gt;,&lt;have_read_length&gt;,&lt;unread_length&gt;</b></p> <p><b>OK</b></p> <p>If there is any error:</p> <p><b>ERROR</b></p> |
| Maximum Response Time                                                                                                                       | /                                                                                                                                                                                                                              |
| Characteristic                                                                                                                              | /                                                                                                                                                                                                                              |

## Parameter

|                                     |                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>&lt;connectID&gt;</b>            | Integer type. Socket service index. Range: 0–11.                                                |
| <b>&lt;read_length&gt;</b>          | Integer type. Maximum length of data to be read. Range: 0–1500. Unit: byte.                     |
| <b>&lt;read_actual_length&gt;</b>   | Integer type. Length of actually retrieved data. Unit: byte.                                    |
| <b>&lt;data&gt;</b>                 | String type. Retrieved data. Unit: byte.                                                        |
| <b>&lt;remoteIP&gt;</b>             | String type. Remote IP address. Valid only when <b>&lt;service_type&gt;</b> is "UDP SERVICE".   |
| <b>&lt;remote_port&gt;</b>          | Integer type. Remote port number. Valid only when <b>&lt;service_type&gt;</b> is "UDP SERVICE". |
| <b>&lt;total_receive_length&gt;</b> | Integer type. Total length of the received data. Unit: byte.                                    |
| <b>&lt;have_read_length&gt;</b>     | Integer type. Length of data that have been retrieved. Unit: byte.                              |
| <b>&lt;unread_length&gt;</b>        | Integer type. Length of data that has not been retrieved. Unit: byte.                           |

### 2.3.10. AT+QISENDEX Send Hex String Data

This command sends hex string data and cannot be applied to "UDP SERVICE" and "TCP LISTENER" sockets.

| AT+QISENDEX Send Hex String Data                                                    |                                                                                                                      |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <p>Test Command</p> <p><b>AT+QISENDEX=?</b></p>                                     | <p>Response</p> <p><b>+QISENDEX: (range of supported &lt;connectID&gt;s),&lt;hex_string&gt;</b></p> <p><b>OK</b></p> |
| <p>Write Command</p> <p><b>AT+QISENDEX=&lt;connectID&gt;,&lt;hex_string&gt;</b></p> | <p>Response</p> <p>(1) If <b>&lt;send_view_mode&gt;=0</b>:</p> <p>If the hex string is sent successfully:</p>        |

|                       |                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | SEND OK                                                                                                                                                                                                                                     |
|                       | If the sending buffer is full:<br>SEND FAIL                                                                                                                                                                                                 |
|                       | If the connection does not exist:<br>ERROR                                                                                                                                                                                                  |
|                       | (2) If <send_view_mode>=1:<br>If the connection has been established:<br>+QISENDEX: <connectID>,<status>,<Freesize><br><br>OK<br><br>If the connection has not been established, abnormally closed, or any parameter is incorrect:<br>ERROR |
| Maximum Response Time | /                                                                                                                                                                                                                                           |
| Characteristic        | /                                                                                                                                                                                                                                           |

## Parameter

|                  |                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------|
| <connectID>      | Integer type. Socket service index. Range: 0–11.                                                                             |
| <hex_string>     | String type. Hex string data. Max. length: 512 bytes.                                                                        |
| <send_view_mode> | Integer type. Enable or disable display <b>AT+QISEND</b> command execution information in URC form.<br>0 Disable<br>1 Enable |
| <status>         | Integer type.<br>0 Send data to the socket buffer successfully<br>1 The receiving buffer is full, send failed                |
| <Freesize>       | Integer type. Free space in the current buffer. Range: 0–10240. Unit: byte.                                                  |

### 2.3.11. AT+QISWTMD Switch Data Access Mode

This command switches the data access modes between buffer access mode, direct push mode and transparent transmission mode. When a socket service is started, the data access mode can be specified via the <access\_mode> of **AT+QIOPEN**. After opening a socket, the data access mode can be changed with **AT+QISWTMD**.

### AT+QISWTMD Switch Data Access Mode

|                                                                          |                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QISWTMD=?</b>                                      | Response<br><b>+QISWTMD:</b> (range of supported <b>&lt;connectID&gt;s</b> ),(range of supported <b>&lt;access_mode&gt;s</b> )<br><br><b>OK</b>                                                                                                                                                                    |
| Write Command<br><b>AT+QISWTMD=&lt;connectID&gt;,&lt;access_mode&gt;</b> | Response<br>If data access mode is switched successfully and <b>&lt;access_mode&gt;</b> is 0 or 1:<br><b>OK</b><br><br>If data access mode is switched successfully and <b>&lt;access_mode&gt;</b> is 2, the module enters the intended data mode:<br><b>CONNECT</b><br><br>If there is any error:<br><b>ERROR</b> |
| Maximum Response Time                                                    | /                                                                                                                                                                                                                                                                                                                  |
| Characteristic                                                           | This command takes effect immediately.<br>The configuration is not saved.                                                                                                                                                                                                                                          |

### Parameter

|                            |                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;connectID&gt;</b>   | Integer type. Socket service index. Range: 0–11.                                                                                    |
| <b>&lt;access_mode&gt;</b> | Integer type. Data access modes of the connection.<br>0 Buffer access mode<br>1 Direct push mode<br>2 Transparent transmission mode |

### 2.3.12. AT+QPING Ping a Remote Server

This command tests the reachability of a host on an Internet protocol network. Before using the ping utility, the host should activate the context of the corresponding **<contextID>** with **AT+QIACT**. The command returns the result within **<timeout>** and the default value of **<timeout>** is 4 s.

### AT+QPING Ping a Remote Server

|                                   |                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QPING=?</b> | Response<br><b>+QPING:</b> (range of supported <b>&lt;contextID&gt;s</b> ), <b>&lt;host&gt;</b> ,(range of supported <b>&lt;timeout&gt;s</b> ),(range of supported <b>&lt;pingnum&gt;s</b> ) |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                     |                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                     | <b>OK</b>                                                                                                                                                                                                                                                                                                                              |
| Write Command<br><b>AT+QPING=&lt;contextID&gt;,&lt;host&gt;[,&lt;timeout&gt;[,&lt;pingnum&gt;]]</b> | Response<br>If a remote host is pinged successfully:<br><b>OK</b><br><br><b>+QPING: &lt;result&gt;[,&lt;IP_address&gt;,&lt;bytes&gt;,&lt;time&gt;,&lt;ttl&gt;][...]</b><br><b>+QPING: &lt;finresult&gt;[,&lt;sent&gt;,&lt;rcvd&gt;,&lt;lost&gt;,&lt;min&gt;,&lt;max&gt;,&lt;avg&gt;]</b><br><br>If there is any error:<br><b>ERROR</b> |
| Maximum Response Time                                                                               | /                                                                                                                                                                                                                                                                                                                                      |
| Characteristic                                                                                      | /                                                                                                                                                                                                                                                                                                                                      |

## Parameter

|                           |                                                                                                                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;contextID&gt;</b>  | Integer type. Context ID. Range: 1–15.                                                                                                                                                                                                                                                  |
| <b>&lt;host&gt;</b>       | String type. Host address in string type. It is a domain name or a dotted decimal IP address.                                                                                                                                                                                           |
| <b>&lt;timeout&gt;</b>    | Integer type. Maximum time to wait for the response of each ping request.<br>Range: 1–255. Default value: 4. Unit: s.                                                                                                                                                                   |
| <b>&lt;pingnum&gt;</b>    | Integer type. Maximum number of ping requests. Range: 1–10. Default value: 4.                                                                                                                                                                                                           |
| <b>&lt;result&gt;</b>     | Integer type. Result of each ping request.<br>0 Received the ping response from the host. In this case, it is followed by<br>,<IP_address>,<bytes>,<time>,<ttl>.<br>Others See <b>Chapter 4</b> for more information.                                                                   |
| <b>&lt;IP_address&gt;</b> | String type. IP address of the remote server formatted as a dotted decimal IP.                                                                                                                                                                                                          |
| <b>&lt;bytes&gt;</b>      | Integer type. The length of each sent ping request. Unit: byte.                                                                                                                                                                                                                         |
| <b>&lt;time&gt;</b>       | Integer type. The time wait for the response of the ping request. Unit: ms.                                                                                                                                                                                                             |
| <b>&lt;ttl&gt;</b>        | Integer type. Time to live value of the response packet for the ping request.                                                                                                                                                                                                           |
| <b>&lt;finresult&gt;</b>  | Integer type. The final result of the command.<br>0 It is finished normally. It is successful to activate the context and find the host. In this case, it is followed by<br>,<sent>,<rcvd>,<lost>,<min>,<max>,<avg>.<br>Others Result codes. See <b>Chapter 4</b> for more information. |
| <b>&lt;sent&gt;</b>       | Integer type. Total number of sent ping requests.                                                                                                                                                                                                                                       |
| <b>&lt;rcvd&gt;</b>       | Integer type. Total number of the ping requests that have received responses.                                                                                                                                                                                                           |
| <b>&lt;lost&gt;</b>       | Integer type. Total number of timed out ping requests.                                                                                                                                                                                                                                  |
| <b>&lt;min&gt;</b>        | Integer type. Minimum response time. Unit: ms.                                                                                                                                                                                                                                          |
| <b>&lt;max&gt;</b>        | Integer type. Maximum response time. Unit: ms.                                                                                                                                                                                                                                          |

<avg> Integer type. Average response time. Unit: ms.

### 2.3.13. AT+QNTF Synchronize Local Time with NTP Server

This command synchronizes the local time with Universal Time Coordinated (UTC) through the NTP server. Before time synchronization, the host should activate the context corresponding to <contextID> with **AT+QIACT**. Depending on the network, it will take max. 125 s to return the result.

| AT+QNTF Synchronize Local Time with NTP Server                                                        |                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test command<br><b>AT+QNTF=?</b>                                                                      | Response<br><b>+QNTF:</b> (range of supported <contextID>s),<server>,(range of supported <port>s),(list of supported <autosettime>s)<br><br><b>OK</b>                    |
| Read command<br><b>AT+QNTF?</b>                                                                       | Response<br>If in the process of local time synchronization:<br><b>+QNTF:</b> <server>,<port><br><br><b>OK</b>                                                           |
| Write command<br><b>AT+QNTF=&lt;contextID&gt;,&lt;server&gt;[,&lt;port&gt;[,&lt;autosettime&gt;]]</b> | Response<br>If the local time is synchronized with NTP server successfully:<br><b>OK</b><br><br><b>+QNTF:</b> <err>,<time><br><br>If there is any error:<br><b>ERROR</b> |
| Maximum Response Time                                                                                 | 125 s, determined by the network.                                                                                                                                        |
| Characteristic                                                                                        | This command takes effect immediately.<br>The configuration is not saved.                                                                                                |

#### Parameter

|               |                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------|
| <contextID>   | Integer type. Context ID. Range: 1–15.                                                            |
| <server>      | String type. NTP server address.                                                                  |
| <port>        | Integer type. NTP server port number. Range: 1–65535. Default value: 123.                         |
| <autosettime> | Integer type. Whether to automatically set synchronized time as local time.<br>0 Not set<br>1 Set |
| <err>         | Integer type. Error codes of the operation. See <b>Chapter 4</b> for more information.            |
| <time>        | String type. Time synchronized with NTP server.                                                   |

The format is "YYYY/MM/DD, hh:mm:ss±zz". Range of zz: -48—+56.

### 2.3.14. AT+QIDNSCFG Configure Address of DNS Server

Before setting the DNS server address, the host must activate the context of corresponding **<contextID>** with **AT+QIACT**.

| <b>AT+QIDNSCFG Configure Address of DNS Server</b>                                              |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test command<br><b>AT+QIDNSCFG=?</b>                                                            | Response<br><b>+QIDNSCFG:</b> (range of supported <b>&lt;contextID&gt;s</b> ), <b>&lt;pridnsaddr&gt;</b> , <b>&lt;secdnsaddr&gt;</b><br><br><b>OK</b>                                                                                                                                                                                                                                                                       |
| Write Command<br><b>AT+QIDNSCFG=&lt;contextID&gt;[,&lt;pridnsaddr&gt;[,&lt;secdnsaddr&gt;]]</b> | Response<br>If the optional parameters are omitted, query the current DNS server addresses of a specified PDP context:<br><b>+QIDNSCFG: &lt;contextID&gt;,&lt;pridnsaddr&gt;,&lt;secdnsaddr&gt;</b><br><br><b>OK</b><br><br>If any of the optional parameters is specified, configure the primary and secondary DNS server addresses of a specified PDP context:<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b> |
| Maximum Response Time                                                                           | /                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Characteristic                                                                                  | This command takes effect immediately.<br>The configurations are not saved.                                                                                                                                                                                                                                                                                                                                                 |

#### Parameter

|                           |                                            |
|---------------------------|--------------------------------------------|
| <b>&lt;contextID&gt;</b>  | Integer type. PDP context ID. Range: 1–15. |
| <b>&lt;pridnsaddr&gt;</b> | String type. Primary DNS server address.   |
| <b>&lt;secdnsaddr&gt;</b> | String type. Secondary DNS server address. |

### 2.3.15. AT+QIDNSGIP Get IP Address by Domain Name

This command gets an IP address by domain name. Before querying the DNS, the host should activate the context of the corresponding **<contextID>** with **AT+QIACT**. Depending on the network, it will return a result in max. 60 s.

| AT+QIDNSGIP Get IP Address by Domain Name                              |                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QIDNSGIP=?</b>                                   | Response<br><b>+QIDNSGIP:</b> (range of supported <contextID>s),<hostname><br><br><b>OK</b>                                                                                                                                                            |
| Write Command<br><b>AT+QIDNSGIP=&lt;contextID&gt;,&lt;hostname&gt;</b> | Response<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b><br><br>The result will be returned as URC.<br><b>+QIURC: "dnsgip",&lt;err&gt;,&lt;IP_count&gt;,&lt;DNS_ttl&gt;</b><br><b>[.....</b><br><b>+QIURC: "dnsgip",&lt;hostIPAddr&gt;]</b> |
| Maximum Response Time                                                  | 60 s, determined by the network.                                                                                                                                                                                                                       |
| Characteristic                                                         | /                                                                                                                                                                                                                                                      |

## Parameter

|              |                                                                       |
|--------------|-----------------------------------------------------------------------|
| <contextID>  | Integer type. Context ID. Range: 1–15.                                |
| <hostname>   | String type. Domain name.                                             |
| <err>        | Integer type. Result code. See <b>Chapter 4</b> for more information. |
| <IP_count>   | Integer type. Number of IP addresses corresponding to the <hostname>. |
| <DNS_ttl>    | Integer type. Time to live of the DNS. Unit: s.                       |
| <hostIPAddr> | String type. IP address of <hostname>.                                |

### 2.3.16. AT+QISDE Control Whether to Echo the Data for AT+QISEND

This command controls whether to echo the data for **AT+QISEND**.

| AT+QISDE Control Whether to Echo the Data for AT+QISEND |                                                                         |
|---------------------------------------------------------|-------------------------------------------------------------------------|
| Test command<br><b>AT+QISDE=?</b>                       | Response<br><b>+QISDE:</b> (list of supported <echo>s)<br><br><b>OK</b> |
| Read command<br><b>AT+QISDE?</b>                        | Response<br><b>+QISDE: &lt;echo&gt;</b><br><br><b>OK</b>                |



|                                               |                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------|
| Write Command<br><b>AT+QISDE=&lt;echo&gt;</b> | Response<br><b>OK</b><br><br>If there is any error:<br><b>ERROR</b>       |
| Maximum Response Time                         | 300 ms                                                                    |
| Characteristic                                | This command takes effect immediately.<br>The configuration is not saved. |

## Parameter

|                     |                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------|
| <b>&lt;echo&gt;</b> | Numeric type. Whether to echo the data for <b>AT+QISEND</b> .<br>0 Do not echo the data<br>1 Echo the data |
|---------------------|------------------------------------------------------------------------------------------------------------|

### 2.3.17. AT+QIGETERROR Query the Error Code of the Last AT Command

If **ERROR** is returned after TCP/IP commands are executed, the detailed information about a result code can be queried with **AT+QIGETERROR**. Please note that **AT+QIGETERROR** only returns the result code of the last TCP/IP AT command.

| <b>AT+QIGETERROR Query the Error Code of the Last AT Command</b> |                                                                                          |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Test command<br><b>AT+QIGETERROR=?</b>                           | Response<br><b>OK</b>                                                                    |
| Execution Command<br><b>AT+QIGETERROR</b>                        | Response<br><b>+QIGETERROR: &lt;err&gt;,&lt;errcode_description&gt;</b><br><br><b>OK</b> |
| Maximum Response Time                                            | 300 ms                                                                                   |
| Characteristic                                                   | /                                                                                        |

## Parameter

|                                    |                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>&lt;err&gt;</b>                 | Integer type. Error codes of the operation. See <b>Chapter 4</b> for more information.                    |
| <b>&lt;errcode_description&gt;</b> | A string parameter indicates the details of error information. See <b>Chapter 4</b> for more information. |

## 2.4. Description of URCs

**+QIURC:** is used at the beginning of the URC of TCP/IP AT commands to be reported to the host. The URC contains the reports about incoming data, closed connection and incoming connection and etc. Actually, there is **<CR><LF>** both before and after URC, but **<CR><LF>** is intentionally not presented.

### 2.4.1. +QIURC: "closed" URC Indicating Connection Closed

When TCP socket service is closed by remote peer or due to a network error, the URC is outputted, and the status of socket service is "closing" (**<socket\_state>=4**). **AT+QICLOSE=<connectID>** can be used to change the **<socket\_state>** to initial.

#### +QIURC: "closed" URC Indicating Connection Closed

**+QIURC: "closed",<connectID>**

Socket service connection is closed.

#### Parameter

|                          |                                                  |
|--------------------------|--------------------------------------------------|
| <b>&lt;connectID&gt;</b> | Integer type. Socket service index. Range: 0–11. |
|--------------------------|--------------------------------------------------|

### 2.4.2. +QIURC: "recv" URC Indicating Incoming Data

In buffer access mode or direct push mode, after receiving data, the module reports a URC to the host.

In buffer access mode, after receiving data, the module reports **+QIURC: "recv",<connectID>** to notify the host. Then host can retrieve data with **AT+QIRD**.

In direct push mode, the received data are outputted to COM port directly.

#### +QIURC: "recv" URC Indicating Incoming Data

**+QIURC: "recv",<connectID>**

Indicates incoming data in buffer access mode. The host can retrieve data via **AT+QIRD**.

**+QIURC: "recv",<connectID>,<currentrecvlength><CR><LF><data>**

Indicates incoming data in direct push mode when the **<service\_type>** is "TCP", "UDP" or "TCP INCOMING".

**+QIURC: "recv",<connectID>,<currentrecvlength>,<remoteIP>,<remote\_port><CR><LF><data>**

Indicates incoming data in direct push mode when **<service\_type>** is "UDP SERVICE".

## Parameter

|                     |                                                  |
|---------------------|--------------------------------------------------|
| <connectID>         | Integer type. Socket service index. Range: 0–11. |
| <currentrecvlength> | Integer type. Length of actually received data.  |
| <remoteIP>          | String type. Remote IP address.                  |
| <remote_port>       | Integer type. Remote port number.                |
| <data>              | Integer type. Received data. Unit: byte.         |

### NOTE

If the buffer is not empty, and the module receives data again, it does not report a new URC until all the received data have been retrieved with **AT+QIRD** from the buffer.

## 2.4.3. +QIURC: "incoming full" Indicate Incoming Connection Reaches the Limit

If the incoming connection reaches the limit, or no socket system resources can be allocated, then the module reports **+QIURC: "incoming full"** for the new incoming connection request.

### +QIURC: "incoming full" Indicate Incoming Connection Reaches the Limit

|                                |                                                                          |
|--------------------------------|--------------------------------------------------------------------------|
| <b>+QIURC: "incoming full"</b> | Indicates that the number of incoming connections has reached the limit. |
|--------------------------------|--------------------------------------------------------------------------|

## 2.4.4. +QIURC: "incoming" Indicate Incoming Connection

If the <service\_type> is "TCP LISTENER", when a remote client connects to this server, the host will automatically assign a free <connectID> for the new connection. The range of <connectID> is 0–11. In such a case, the module will report the URC. The <service\_type> of the new connection will be "TCP INCOMING", and the <access\_mode> will be buffer access mode.

### +QIURC: "incoming" Indicate Incoming Connection

|                                                                                                             |                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>+QIURC:</b><br><b>"incoming",&lt;connectID&gt;,&lt;serverID&gt;,&lt;remoteIP&gt;,&lt;remote_port&gt;</b> | When the new incoming connection is accepted by <serverID>, the allocated <connectID>, <remoteIP> and <remote_port> will be informed by this URC. |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

## Parameter

|             |                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <connectID> | Integer type. Assign this socket service for the incoming connection, which is automatically specified by the module. Range: 0–11.          |
| <serverID>  | Integer type. The incoming <connectID> accepted by the server whose <service_type> is "TCP LISTENER" and listening socket ID is <serverID>. |
| <remoteIP>  | String type. Remote IP address of the incoming <connectID>.                                                                                 |

---

**<remote\_port>** Integer type. Remote port of the incoming **<connectID>**.

---

#### 2.4.5. +QIURC: "pdpdeact" Indicate PDP Deactivation

PDP context may be deactivated by the network. The module will report the URC to the host about PDP deactivation. In such a case, the host must execute **AT+QIDEACT** to deactivate the context and reset all connections.

##### +QIURC: "pdpdeact" Indicate PDP Deactivation

|                                             |                                                  |
|---------------------------------------------|--------------------------------------------------|
| <b>+QIURC: "pdpdeact",&lt;contextID&gt;</b> | <b>&lt;contextID&gt;</b> context is deactivated. |
|---------------------------------------------|--------------------------------------------------|

---

#### Parameter

---

**<contextID>** Integer type. Context ID. Range: 1–15.

---

#### 2.4.6. +QIURC: "wakeup/data" Indicate Wake-up Packet Data Receive

This URC is reported when the module receives the wake-up packet data sent by the server.

##### +QIURC: "wakeup/data" Indicate Wake-up Packet Data Receive

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| <b>+QIURC: "wakeup/data",&lt;connectID&gt;</b> | Wake-up data is sent by the server. |
|------------------------------------------------|-------------------------------------|

---

#### Parameter

---

**<connectID>** Integer type. Context ID. Range: 1–11.

---

# 3 Examples

## 3.1. Configure and Activate a Context

### 3.1.1. Configure a Context

```
AT+QICSGP=1,1,"UNINET","", "",1 //Configure context 1. China Unicom APN: "UNINET".
OK
```

### 3.1.2. Activate a Context

```
AT+QIACT=1 //Activate context 1. Depending on the network, the maximum
            response time is 150 s.
OK //Activated the context successfully.
AT+QIACT? //Query the context state.
+QIACT: 1,1,1,"10.7.157.1"
OK
```

### 3.1.3. Deactivate a Context

```
AT+QIDEACT=1 //Deactivate context 1.
OK //Deactivated the context successfully. Depending on the
    network, the maximum response time is 40 s.
```

## 3.2. TCP Client Works in Buffer Access Mode

### 3.2.1. Set up a TCP Client Connection and Enter Buffer Access Mode

```
AT+QIOPEN=1,0,"TCP","220.180.239.212",8009,0,0 //Context is 1 and <connectID> is 0. Before
                                                    using AT+QIOPEN, the host should activate
                                                    the context with AT+QIACT first.
```

OK

+QIOPEN: 0,0

//TCP client connected successfully. It is suggested to wait for 150 s for the URC **+QIOPEN: <connectID>,<err>**. If the URC cannot be received in 150 s, the host could use **AT+QICLOSE** to close the socket.

**AT+QISTATE=1,0**

//Query the connection status of socket service 1.

+QISTATE: 0,"TCP","220.180.239.212",8009,65514,2,1,0,0,"usbmodem"

OK

### 3.2.2. Send Data in Buffer Access Mode

**AT+QISEND=0**

//Send variable-length data.

> test1<ctrl+Z>

SEND OK

//**SEND OK** does not mean the data has been sent to the server successfully. The host can query whether the data has reached the server via **AT+QISEND=0,0**.

**AT+QISEND=0,4**

//Send fixed-length data and the data length is 4 bytes.

> test

SEND OK

**AT+QISEND=0,0**

//Query the length of sent data.

+QISEND: 9,9,0

OK

**AT+QISENDEX=0,"3132333435"**

//Send Hex string data.

SEND OK

**AT+QISEND=0,0**

//Query the length of sent data, acknowledged data and unacknowledged data.

+QISEND: 14,14,0

OK

### 3.2.3. Retrieve Data from Remote Server in Buffer Access Mode

+QIURC: "recv",0

//The received data when **<connectID>=0**.

**AT+QIRD=0,1500**

//Read data, the maximum length of data to be retrieved is 1500 bytes.

+QIRD: 14

//The length of actually received data is 14 bytes.

test1

```

OK
AT+QIRD=0,1500
+QIRD: 0 //No data in buffer.

OK
AT+QIRD=0,0 //Query the total length of received data, including read and unread data.
+QIRD: 14,14,0

OK

```

### 3.2.4. Close a Connection

```

AT+QICLOSE=0 //Close a connection whose <connectID> is 0. Depending on the
              network, the maximum response time is 10 s.

OK

```

## 3.3. TCP Client Works in Transparent Access Mode

### 3.3.1. Set up a TCP Client Connection and Enter Transparent Access Mode

```

AT+QIOPEN=1,0,"TCP","220.180.239.212",8009,0,2 //Context is 1 and <connectID> is 0. Before using
AT+QIOPEN, the host should activate the
context with AT+QIACT.

CONNECT //TCP client connected successfully. It is
suggested to wait for 150 s for the URC
CONNECT. If the URC cannot be received in
150 s, the host could use AT+QICLOSE to close
the socket.

```

### 3.3.2. Send Data in Transparent Access Mode

```

<All data got from COM port will be sent to internet directly>

```

### 3.3.3. Retrieve Data from Remote Server in Transparent Access Mode

```

Test 1 //All data received from the Internet are outputted
        directly via COM port.

```

### 3.3.4. Close a TCP Client

**AT+QICLOSE=0**

//After using **+++** to exit transparent transmission mode, the host could use **AT+QICLOSE** to close the TCP link. Depending on the network, the maximum response time is 10 s.

OK

## 3.4. TCP Client Works in Direct Push Mode

### 3.4.1. Set up a TCP Client Connection and Enter Direct Push Mode

**AT+QIOPEN=1,0,"TCP","220.180.239.212",8009,0,1** //Context is 1 and **<connectID>** is 0. Before using **AT+QIOPEN**, the host should activate the context with **AT+QIACT**.

OK

**+QIOPEN: 0,0**

//TCP client is connected successfully. It is suggested to wait for 150 s for the URC **+QIOPEN: <connectID>,<err>**. If the URC cannot be received in 150 s, the host could use **AT+QICLOSE** to close the socket.

**AT+QISTATE=1,0**

//Query if the connection state of **<connectID>** is 0.

**+QISTATE: 0,"TCP","220.180.239.212",8009,65344,2,1,0,1,"usbmodem"**

OK

### 3.4.2. Send Data in Direct Push Mode

**AT+QISEND=0**

//Send variable-length data.

**> test1<ctrl+Z>**

**SEND OK**

//**SEND OK** does not mean the data have been received by the server successfully. Host can query whether the data have reached the server with **AT+QISEND=0,0**.

**AT+QISEND=0,5**

//Send fixed length data and the data length is 5 bytes.

**> test2**

**SEND OK**

**AT+QISEND=0,0**

//Query the lengths of sent, acknowledged and unacknowledged data.



```
+QISEND: 10,10,0 //A total of 10 bytes of data have been sent, and all 10
                    bytes have been acknowledged.
```

```
OK
```

### 3.4.3. Retrieve Data from Remote Server in Direct Push Mode

```
+QIURC: "recv",0,4 //Retrieve data from remote server.
test
```

### 3.4.4. Close a TCP Client

```
AT+QICLOSE=0 //Close the connection whose <connectID> is 0.
              Depending on the network, the maximum response time
              is 10 s.
```

```
OK
```

## 3.5. TCP Server Works in Buffer Access Mode

### 3.5.1. Start a TCP Server

```
AT+QIOPEN=1,1,"TCP LISTENER","127.0.0.1",0,2020,0 //Context is 1 and <connectID> is 1. Before
                                                    using AT+QIOPEN, the host should activate
                                                    the context with AT+QIACT.
```

```
OK
```

```
+QIOPEN: 1,0 //TCP server is opened successfully.
AT+QISTATE=0,1 //Query whether the connection status of
               <contextID> is 1.
```

```
+QISTATE: 1,"TCP LISTENER","10.7.157.1",0,2020,3,1,1,0,"usbmodem"
```

```
OK
```

### 3.5.2. Accept TCP Incoming Connection

```
+QIURC: "incoming",11,1,"172.31.242.222",54091 //Accepted a new TCP connection. The
                                                    <service_type> is "TCP incoming", and
                                                    <connectID> is 11.
```

### 3.5.3. Retrieve Data from Incoming Connection

```
+QIURC: "recv",11 //Received data from remote incoming connection.
AT+QIRD=11,1500 //Retrieve data received from incoming connection.
+QIRD: 4 //Length of actually retrieved data is 4 bytes.
test

OK
AT+QIRD=11,1500
+QIRD: 0 //No data in buffer.

OK
AT+QIRD=11,0 //Query the total length of received data, including read and unread data.
+QIRD: 4,4,0

OK
```

### 3.5.4. Close a TCP Server

```
AT+QICLOSE=1 //Close the incoming connection. Depending on the network, the
//maximum response time is 10 s by default.

OK
AT+QICLOSE=1 //Close TCP server listening.

OK
```

## 3.6. UDP Service

### 3.6.1. Start a UDP Service

```
AT+QIOPEN=1,2,"UDP SERVICE","127.0.0.1",0,3030,0 //Start a UDP service whose <connectID> is
//2 and <contextID> is 1. Before using
//AT+QIOPEN, the host should activate the
//context with AT+QIACT.

OK

+QIOPEN: 2,0 //UDP service is opened successfully.
AT+QISTATE=0,1 //Query if the connection status of
//<contextID> is 1.

+QISTATE: 2,"UDP SERVICE","10.7.157.1",0,3030,2,1,2,0,"usbmodem"
```

OK

### 3.6.2. Send UDP Data to Remote Client

|                                         |                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>AT+QISEND=2,10,"10.7.89.10",6969</b> | //Send 10-byte long data to a remote client whose IP is 10.7.89.10 and the remote port is 6969. |
| <br>>1234567890                         |                                                                                                 |
| SEND OK                                 |                                                                                                 |

### 3.6.3. Receive Data from Remote Client

|                                         |                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>+QIURC: "recv",2</b>                 | //Receive data from remote client.                                                                        |
| <b>AT+QIRD=2</b>                        | //Retrieve UDP data. One whole UDP packet will be outputted. There is no need to specify the read length. |
| <b>+QIRD: 4,"10.7.76.34",7687</b>       | //The retrieved data length is 4. The remote IP address is 10.7.76.34 and remote port is 7687.            |
| AAAA                                    |                                                                                                           |
| OK                                      |                                                                                                           |
| <b>AT+QIRD=2</b>                        | //Retrieve data.                                                                                          |
| <b>+QIRD: 0</b>                         | //No data in buffer.                                                                                      |
| OK                                      |                                                                                                           |
| <b>AT+QISEND=2,10,"10.7.76.34",7687</b> | //Send data to the remote client whose IP is 10.7.76.34 and remote port is 7687.                          |
| >1234567890                             |                                                                                                           |
| SEND OK                                 |                                                                                                           |

### 3.6.4. Close a UDP Service

|                     |                      |
|---------------------|----------------------|
| <b>AT+QICLOSE=2</b> | //Close the service. |
| OK                  |                      |

## 3.7. PING

|                                   |                                                                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AT+QPING=1,"www.baidu.com"</b> | //Ping <a href="http://www.baidu.com">www.baidu.com</a> in context 1. Before pinging the destination IP address, the host should activate the context with |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

AT+QIACT.

OK

+QPING: 0,"61.135.169.125",32,192,255

+QPING: 0,"61.135.169.125",32,240,255

+QPING: 0,"61.135.169.125",32,241,255

+QPING: 0,"61.135.169.125",32,479,255

+QPING: 0,4,4,0,192,479,288

### 3.8. Synchronize Local Time

AT+QNTP=1,"202.112.10.36",123

//Synchronize local time with NTP server "202.112.10.36:123".  
Before synchronizing the time, the host should activate the context with **AT+QIACT**.

OK

+QNTP: 0,"2019/07/21,06:10:59+00"

AT+CCLK?

+CCLK: "19/07/21,06:11:05+00"

OK

### 3.9. Get Last Error Code

AT+QIOPEN=1,"TCP","220.180.239.212",8009,0,1 //Send **AT+QIOPEN** with missing <connectID>.

ERROR

AT+QIGETERROR

+QIGETERROR: 552,invalid parameters

OK

### 3.10. Optional Configurations of TCP

```
//Set wake-up packet data.
AT+QICFG="wakeup/data",1                                     //Query the current setting.
+QICFG: "wakeup/data",1,"",0

OK
AT+QICFG="wakeup/data",1,"5155454377616b657570"             //Set wake-up packet data.
OK
AT+QICFG="wakeup/data",1                                     //Query the current wake-up packet data.
+QICFG: "wakeup/data",1,"5155454377616b657570",0

OK
AT+QICFG="wakeup/data",1,""                                   //Clear and disable wake-up feature.
OK
AT+QICFG="wakeup/data",1                                     //Query the current setting after the wake-up feature is disabled.
+QICFG: "wakeup/data",1,"",0

OK
AT+QICFG="wakeup/data",1,"quec wakeup",2                     //Configure wake-up packet data to string
OK                                                            type.

//Query or set to send heartbeat packet data periodically.
AT+QICFG="send/auto",0,30,"313233343536",0,0                //Configure heartbeat packet data to
OK                                                            hexadecimal ASCII code.

AT+QICFG="send/auto",0                                       //Query the current setting.
+QICFG: "send/auto",0,30,"123456",0,0

OK
AT+QICFG="send/auto",0,30,"hi quectel",2,0                  //Configure heartbeat packet data to string type.
OK
AT+QICFG="send/auto",0                                       //Query the current setting.
+QICFG: "send/auto",0,30,"hi quectel",2,0

OK

//Query or set to filter the specified data.
AT+QICFG="recv/ignore",0,"313233343536",0                    //Configure filtered data to hexadecimal ASCII code.
OK
AT+QICFG="recv/ignore",0                                     //Query the current setting.
+QICFG: "recv/ignore",0,"123456",0
```

OK

**AT+QICFG="recv/ignore",0,"313233343536",1**

//Configure filtered data to hexadecimal string.

OK

**AT+QICFG="recv/ignore",0**

//Query the current setting.

**+QICFG: "recv/ignore",0,"313233343536",1**

OK

# 4 Summary of Error Codes

If **ERROR** is returned after executing TCP/IP AT commands, the detailed information about errors can be queried with **AT+QIGETERROR**. Please note that **AT+QIGETERROR** just returns the result code of the last TCP/IP AT command.

**Table 3: Summary of Error Codes**

| <err> | <errcode_description>         |
|-------|-------------------------------|
| 0     | Operate successfully          |
| 550   | Unknown error                 |
| 551   | Operation blocked             |
| 552   | Invalid parameters            |
| 553   | Memory allocation failed      |
| 554   | Socket creation failed        |
| 555   | Operation not supported       |
| 556   | Socket bind failed            |
| 557   | Socket listen failed          |
| 558   | Socket write failed           |
| 559   | Socket read failed            |
| 560   | Socket accept failed          |
| 561   | Activate PDP context failed   |
| 562   | Deactivate PDP context failed |
| 563   | Socket identity has been used |
| 564   | DNS busy                      |

---

|     |                         |
|-----|-------------------------|
| 565 | DNS parse failed        |
| 566 | Socket connect failed   |
| 567 | Connection reset        |
| 568 | System busy             |
| 569 | Operation timeout       |
| 570 | PDP context deactivated |
| 571 | Cancel sending          |
| 572 | Operation not allowed   |
| 573 | APN not configured      |
| 574 | Port busy               |

---



# 5 Appendix References

**Table 4: Related Document**

| Document Name                                              |
|------------------------------------------------------------|
| [1] Quectel_ECx00M&ECx00N&EC800K_Series_AT_Commands_Manual |

**Table 5: Terms and Abbreviations**

| Abbreviation | Description                                        |
|--------------|----------------------------------------------------|
| 3GPP         | 3rd Generation Partnership Project                 |
| ACK          | Acknowledgement                                    |
| APN          | Access Point Name                                  |
| ASCII        | American Standard Code for Information Interchange |
| CHAP         | Challenge Handshake Authentication Protocol        |
| CS           | Circuit Switching                                  |
| DNS          | Domain Name System                                 |
| FIN          | Finish                                             |
| ID           | Mostly refers to Identifier in terms of software   |
| IP           | Internet Protocol                                  |
| NTP          | Network Time Protocol                              |
| NV           | Non-Volatile                                       |
| PAP          | Password Authentication Protocol I                 |
| PDP          | Packet Data Protocol                               |
| PPP          | Point-to-Point Protocol                            |

---

|        |                                                 |
|--------|-------------------------------------------------|
| PS     | Packet Switching                                |
| QoS    | Quality of Service                              |
| TCP    | Transmission Control Protocol                   |
| TCP/IP | Transmission Control Protocol/Internet Protocol |
| UART   | Universal Asynchronous Receiver& Transmitter    |
| UDP    | User Datagram Protocol                          |
| URC    | Unsolicited Result Code                         |
| USB    | Universal Serial Bus                            |
| (U)SIM | (Universal) Subscriber Identity Module          |
| UTC    | Coordinated Universal Time                      |

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